

Intel Motherboards						
Brand Model No.	ASUS P5Q-EM	ECS GF9300T-A	Gigabyte EG43M-S2H	MSI G43M2	MSI P7NGM DIGITAL	ZOTAC GeForce 9300
						
Chipset	Intel G45	NVIDIA GeForce 9300	Intel G43	Intel G43	NVIDIA GeForce 9300	NVIDIA GeForce 9300
Socket	LGA 775	LGA 775	LGA 775	LGA 775	LGA 775	LGA 775
Northbridge	Intel G45	MCP7I	Intel G43	Intel G43	MCP7A	MCP7A
Southbridge	Intel ICH10R		Intel ICH10	Intel ICH10		
Price (Rs)	8,375	7,060	7,500	5,500	6,700	5,999
+	Quality board	Good overall performance	Good BIOS for overclockers	Inexpensive	Good overall performance	Good value for money
-	Expensive	Slightly expensive	Slightly expensive	Cramped layout	Not for overclockers	BIOS lacks overclocking features
Grand Totals (Out Of 100)	60.00	58.60	62.88	48.67	64.29	59.87
Features (Out Of 70)	41.20	34.09	42.95	28.88	40.99	35.14
Performance (Out Of 30)	18.80	24.51	19.94	19.79	23.31	24.73
Features						
Front Side Bus Speed	1600 MHz	1333 MHz	1333 MHz (1600 MHz OC)	1333 MHz	1333 MHz	1333 MHz
Memory Type / Max Freq. Supported	DDR2 / 800 MHz (1066 MHz OC)	DDR2 / 800 MHz	DDR2 / 1066 MHz (1200 MHz OC)	DDR2 / 800 MHz	DDR2 / 800 MHz	DDR2 / 800 MHz
Memory Supported / No. Of slots	16 GB / 4	8 GB / 4	8 GB / 4	4 GB / 2	8 GB / 4	16 GB / 4
Power Saving features (with GPU)	✗, EPU	✓	✗, DES	✗	✓	✓
No. Of SATA / PATA Ports	6 / 1	6 / 1	5 / 1	4 / 1	6 / 1	6 / 1
No. Of Rear E-SATA Ports	0	0	1	0	0	0
HDMI / DVI / VGA	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✗ / ✗ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓
Multi GPU Support (SLI / CrossFire / N)	✗	✗	✗	✗	✗	✗
Audio Controller / Channels / HD Compliant (Y / N)	ALC1200 / 8 / ✓	ALC889 / 8 / ✓	ALC888 / 8 / ✓	ALC888 / 8 / ✓	ALC888 / 8 / ✓	ALC888 / 8 / ✓
S / PDIF / Optical Port	✓ / ✗	✗ / ✗	✓ / ✓	✗ / ✗	✓ / ✗	✓ / ✗
No Of LAN Ports / 10 / 100 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit
Integrated WiFi (Y / N)	✗	✗	✗	✗	✗	✗
No. Of Rear USB / FireWire Ports	6 / 1	6 / 0	6 / 1	4 / 1	4 / 1	4 / 0
All Solid State Capacitors (Y / N)	✓	✗	✓	✗	✓	✗
Overclocking Options						
CPU / FSB / Memory	✓ / ✓ / ✓	✗ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✗ / ✓	✓ / ✓ / ✓	✗ / ✓ / ✓
Overvolting (CPU / Memory / FSB / PCIe / Northbridge)	✓ / ✓ / ✓ / ✓ / ✓	✗ / ✓ / ✗ / ✗ / ✓	✓ / ✓ / ✓ / ✓ / ✓	✗ / ✓ / ✓ / ✗ / ✗	✓ / ✓ / ✓ / ✗ / ✗	✗ / ✓ / ✗ / ✗ / ✓
Package Bundle						
No of SATA / PATA Cables Provided	3 / 1	4 / 1	2 / 1	1 / 1	1 / 1	1 / 1
Performance						
PC Mark 2005						
CPU	9006	9618	9625	9650	8991	10112
Memory / Graphics	5950 / 1886	5656 / 2361	5933 / 1819	6083 / 1767	5708 / 2377	5795 / 2300
HDD	6755	6680	6609	6692	6110	6795
Overall	5663	6343	6181	6157	5802	6418
SiSoft Sandra 2008						
CPU Arithmetic (Dhrystone / Whetstone)	37053 / 29486	55563 / 43912	55325 / 44064	55707 / 44270	37175 / 29589	57890 / 46319
CPU Multimedia (Integer / Floating)	221990 / 120954	327057 / 179056	330038 / 180734	330759 / 181133	222010 / 120491	345168 / 189008
Drive Index (MBps) / Random Access Time (ms)	76 / 13	77 / 13	77 / 13	77 / 13	76 / 12	77 / 14
Memory Bandwidth Score (Integer / Floating)	5364 / 5353	4880 / 4850	5316 / 5301	5496 / 5497	4967 / 4956	5049 / 5042
3D Mark 2005:						
3D Marks / CPU Score	1583 / 4627	2810 / 10147	1478 / 5091	1494 / 4709	2684 / 9732	2824 / 10319
Video Encoding (DivX 6.2) Time	68	70	69	68	68	67
File Transfer - 4 GB File	188	188	188	192	206	193
Doom 3 (640x480 / 1024x768 - ultra no AA/AF)	21.0 / 15.8	57.4 / 32.0	27.2 / 15.9	27.8 / 15.7	56.2 / 32.4	56.8 / 31.5
Company of Heroes (800x600/1280x1024)	12.2 / 5.7	21.2 / 12.8	11.5 / 7	11.6 / 5.9	19.5 / 13.1	20.5 / 13.2

In the games – *Doom 3* and *Company of Heroes* – all the GeForce 9300 boards from ECS, MSI and ZOTAC scored similarly. *Company Heroes* is an RTS and 20 fps in a game like that can’t be called great, but it’s still close to twice as much as the Intel G45 and G43 boards from MSI and ASUS could achieve. *Doom 3* painted a different picture. At 640 x 480 with ultra quality but no anti-aliasing, the game runs pretty comfortably at about 50 fps. With the resolution set to

1024 x 768, frame rates dropped to around 30 fps. When *Doom 3* was first released, the best of cards were only able to run the game at around 40 to 50 fps at 1024 x 768. In that aspect, motherboard chipsets have come a long way since then. The MSI G43M2 and the G45 powered ASUS P5Q-EM could run *Doom 3* but scored close to half of what the 9300s were capable of. 3D Mark 2005, another benchmark that stresses a lot on the on the graphics subsystem,

ran very well on the ECS GF9300T-A and the Zotac GeForce 9300. They scored 2810 and 2824 respectively. What was interesting to see in 3D Mark tests was that the NVIDIA boards helped get the CPU score up a lot as compared to the Intel based ones. Intel’s onboard graphics solutions seem to depend a lot on the CPU. **Value** If we were look at just price, the MSI G43M2 is the cheapest of the bunch at Rs 5,500. If you did want something really cheap and basic, then this would be a decent buy. Using this as a base to build a powerful system is honestly not a bad idea. The board itself is limited in features and the design



isn’t exciting either. The price of Rs 5,500 is still fairly high for this board. For the same price, it’s easy to find motherboards of previous generations from Intel or even NVIDIA. The best bang for your buck board is the ZOTAC GeForce 9300. For just Rs 500 more than the MSI G43M2, you get a really decent board with good expandibility options. Even though this is a great board for everyone to buy, it’s not for hardcore enthusiasts who want to push the board to its limits, and squeeze every bit of performance out of the processor. It makes good sense for an HTPC, where a compact design and good HD playback performance is required. The GeForce 9300’s IGP is easily able to handle HD playback as well as

high definition. The fan on the heatsink will make sure things remain stable and when you feel like playing a few simple games, you can do that too, as this is the best performing board for Intel-based, onboard graphic systems. If performance doesn’t bother you, and you have a real graphic card to do that for you, then the Gigabyte EG43M-S2H can be the next best option for you. It has all the features of the ASUS P5Q-EM, including all the overclockability functions. Unlike the MSI board, the EG43M-S2H is slightly more open and well laid out. The ASUS P5Q-EM is a good board but the only problem with it is its price. At Rs 8,375, it’s the most expensive one. The onboard performance is nowhere close to the 9300 chipsets, and it only makes sense as a board for overclockers who might, for some odd reason, not want a dedicated graphics card. The ECS GF9300T-A board is a little bit more of an enthusiast

level board. With its three PCI slots, there is loads of space for adding additional cards. This is easily one of those boards you buy for now, and add a dedicated graphics down the line – just don’t expect to put in additional cards to SLI them. **Motherboards For AMD Processors** Unlike the Intel motherboard chipsets, there are many more AMD chipsets with onboard graphics in the market and we received loads of them. The ones we tested were mainly the ones with the AMD 780 and 790 chipsets as well as the NVIDIA GeForce 8000 series and the NVIDIA 780a chipset. Unlike the chipsets for Intel’s platform, there are boards that support NVIDIA’s SLI and AMD’s CrossFire. The price range of the products in this category range from as low as Rs 4,000 to as much as Rs 15,500. There were also a few more brands in this category than the previous one.



Simply Believe

NO Server Setup Required
Easy connection to Linux, Windows, Citrix, Goglobal etc. Servers

Embedded Openoffice Suite, Email Client (Thunderbird), Web browser (Firefox), Media Player (Audio/Video), PDF Reader, Messengers, Skype

Embedded DOS Client (connect to any dos folder on Linux, Novel, Windows, Server) - (Full Screen Dos, Switchable session)

Multi-Session, Multi-Protocol Multi-server support

Local Web Browser (with JVM 1.6, flash plugins, media codecs)

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